



General Public Behaviour & Attitude towards Road Safety

Table of Contents

1. INTRODUCTION.....	3
2. DATA ANALYTICS.....	3
2.1 TYPES OF DATA.....	3
2.1.1 <i>Nominal Data</i>	3
2.1.2 <i>Ordinal Data</i>	4
2.1.3 <i>Interval/Ratio</i>	4
2.2 TYPES OF DATA ANALYTICS	5
2.2.1 <i>Descriptive Analytics</i>	5
2.2.2 <i>Diagnostics Analytics</i>	5
2.2.3 <i>Predictive Analysis</i>	5
2.2.4 <i>Prescriptive Analysis</i>	6
3. IMPACT OF ROAD SAFETY ON DIFFERENT BUSINESS INDUSTRIES	6
3.1 PROBLEM 1: OVER-SPEEDING INFLUENCE ON INSURANCE PROVIDERS	7
3.2 PROBLEM 2: SEAT BELT UTILIZATION – MANUFACTURING.....	8
3.3 PROBLEM 3: DRIVER DISTRACTION – AUTOMOBILE TECHNOLOGY INVENTOR (RESEARCH & DEVELOPMENT)	9
4. CONCLUSION	10
5. REFERENCE	12

1. Introduction

Road transportation is the most intricate and hazardous system that individuals have to deal with on a daily basis (WHO, 2004). As per Gitelman et al., (2012), more than 3000 people die daily in road accidents, while the global road accident financial cost is estimated at more than 518 billion dollars annually. The data makes it equally crucial to comprehend the general public's attitudes and actions. Despite introducing various laws and regulation management worldwide, road safety has remained a universal concern, and issues still affect road users (Gitelman et al., 2012). This report uses contemporary literature to assess and articulate the public's behaviour and attitude towards road safety. Section 2 captures the Level of measurement in Data Analytics by categorizing the data as Nominal, Ordinal, Interval/Ratio, and different types of data analytics. Section 3 discusses three critical problems in three different contexts and the appropriate data analytics method for making a prescriptive and informed decision. Section 4 summarizes the conclusion.

2. Data Analytics

Data analytics is "the application of methods, tools, and models to analyze data to gain insight and make informed decisions" (Raghupathi et al., 2021). According to Forbes (2020), the total amount of data generated by the world in 2010 was two zettabytes (ZB). With increasing company digitalization and transformation, overall data generation is forecasted to exceed 44 zettabytes (ZB) by 2020 (World Economic Forum, 2019).

Due to the high rate of reported incidents, data analytics has assisted governments, automobile and Insurance companies worldwide to emphasize security and well-being (Bagloee et al., 2016). As per Bagloee et al., (2016), automobile manufacturers heavily invest in research and development, using big data analytics to analyze the high magnitude of data to forecast future events. The types of data and analytics have been covered in the section below.

2.1 Types of Data

The data type is critical to analyze the level of measurement of each dataset as listed below:

2.1.1 Nominal Data

In analytical terminology, nominal data is used to identify variables but has no quantitative value (CFI, 2023). It is the most basic type of scale of measurement. Based on the selected

topic, Nominal variables can be identified as Car registration number, region in New Zealand, occupation, ethnicity, and driving license type.

Level of measurement – Frequencies and Proportions (Freeman, 2011).

Variable				
Nominal Data				
Car Registration Number	Region	Occupation	Ethnicity	License Type
UKL989	Auckland	White Collar	NZ Maori	Full
PBM845	Northland	Blue Collar	Asian	Learner
KMN775	Waikato	Retired	European	Restricted

Figure 1: Nominal Data Example

2.1.2 Ordinal Data

Ordinal data refers to data categorization into groups that have a natural rank order within a variable (Bhandari, 2020). However, Bhandari (2020) also states that the variations in distance between the categories are irregular or undetermined. Below is a suitable example, considering the behavioural aspect towards road safety.

Level of measurement – Frequencies and Proportions (sometimes Mean) (Freeman, 2011).

Variable	
Ordinal Data	
Enforcing the speed limit helps to reduce road deaths?	Strongly disagree
	Disagree
	Neutral
	Agree
	Strongly agree
	Don't know

Figure 2: Ordinal Data Example

2.1.3 Interval/Ratio

Interval/Ratio data helps measure variables with uniform intervals. This kind of measurement is frequently applied to make accurate computations and comparisons. The difference between Interval and Ratio data is that Interval data holds no true zero. Vehicle speed is an example of interval data because a distance of 20 to 30 kilometres is equivalent to 60 to 70

kilometres. Personal income is a suitable example for ratio data since it has a true zero point that enables meaningful comparison.

Level of measurement – Mean, Median and Standard Deviation (Freeman, 2011).

Variable			
Interval Data		Ratio Data	
Vehicle Speed	20-30km/h	Personal Income	0-30K
	30-40km/h		31k-70k
	40-50km/h		70k & above

Figure 3: Interval & Ratio Data Example

2.2 Types of Data Analytics

Data analytics enables leadership and management to make informed decisions for the organization's success. The four major types of data analytics are discussed below.

2.2.1 Descriptive Analytics

Descriptive analytics, the most straightforward and widely employed data analysis, is pivotal in transforming vast datasets into practical and manageable subsets (Black, 2023). Its primary objective is to identify historical data trends, answering the fundamental questions of "What and how it happened." This foundational analysis serves as a crucial base for other advanced forms of analytics (Harvard Business School, 2021).

2.2.2 Diagnostics Analytics

The subsequent logical study, "Why did this happen?" is answered by diagnostic analytics (Harvard Business School, 2021). Its primary purpose is to uncover the unrecognised issue's underlying causes and identify previously undiscovered correlations and movements between variables (Gartner, 2023). It is the most critical and detailed type of data analytics, as once the root cause has been identified, other data analytics help assess the future impact and how risk can be mitigated.

2.2.3 Predictive Analysis

As its name suggests, predictive analysis is vital in forecasting future events and trends, helping organizations answer the crucial question of "What will happen in the future?" (Harvard

Business School, 2021). Applying sophisticated statistical techniques to historical data empowers companies to comprehend potential outcomes and devise strategies aligned with favourable results while mitigating risks (ProjectPro, 2023).

2.2.4 Prescriptive Analysis

Prescriptive analytics offers insightful recommendations to businesses, guiding them towards actions that can maximize critical Key Performance Indicators (KPIs) (Fallon, 2023). By employing optimization and simulation techniques, Prescriptive analytics addresses the fundamental question of "What should a business do?" (Raghupathi et al., 2021). Fallon (2023) emphasizes that prescriptive analysis adopts a highly technical approach, utilizing machine learning, Artificial Intelligence, and advanced mathematical algorithms to gain profound insights into future outcomes. According to Harvard Business School (2021), manual analysis can yield feasible future actions; however, prescriptive analytics, powered by cutting-edge technology, provides recommendations that might be challenging to implement due to their complexity.

To harness the full potential of data analytics, the data must undergo thorough cleansing and fix structural errors, manage missing data and validate, setting the stage for the system's technological capabilities to analyse the data efficiently (Tableau, 2023).

The following section identifies three problems and applications of data analytics.

3. Impact of Road safety on different Business Industries

This section addresses three critical problems concerning the public's attitude and its impact on various contexts through the application of data analytics. By leveraging data-driven insights, businesses can effectively assess these challenges and make informed, strategic decisions for the future. The careful analysis of public attitudes will provide valuable guidance in navigating these complex issues and pave the way for more robust and successful outcomes across different sectors. Each problem identifies the variable and data analysis type essential for conducting a comprehensive analysis of a driver's behaviour and attitude concerning Speeding, Seat belt utilization and Distraction while driving.

3.1 Problem 1: Over-Speeding Influence on Insurance Providers

Speeding remains a significant challenge faced by Transport Agencies worldwide, leading to tragic annual fatalities and accidents (NZ Transport Agency, 2018). Despite implementing various road safety awareness programs and stringent penalties enforced by authorities, many drivers persist in neglecting speed restrictions, endangering their lives and those of others (Gitelman et al., 2012). In response to the critical road safety issue, the New Zealand government has taken proactive measures by installing a comprehensive network of speed cameras, including mobile and static cameras, to cover the entire road network (NZ Police, 2023). It has also been argued whether the data collected through these cameras depict a fair image of the offence, and NZ Police (2023) shares a code of conduct for such data like the images and other relevant information, is securely and digitally stored, ensuring its integrity and making it immune to tampering at any given time.

Since the reduction of the speed tolerance limit from 10km/h to 4km/h over the speed limit in 2010, there has been a notable increase in the number of speeding tickets issued to date (NZ Herald, 2023). According to Insurance Business (2020), an astonishing \$272 million worth of fines were accumulated in New Zealand between 2019-20, with speeding violations topping the list as the most prevalent offence; over two million motorists were caught speeding, resulting in penalties amounting to more than \$240 million. However, due to the substantial rise in speeding fines, drivers have become hesitant to report such offences to their insurance providers (NZ Herald, 2023). This situation poses a higher liability for insurance companies, as historical driving records are crucial in risk assessment and premium quotations (Forbes, 2023). In light of this, the Insurance Council of New Zealand (ICNZ) has established a duty of disclosure for insurance applicants, stressing the importance of honesty and accuracy in providing information to insurers (ICNZ, 2023). Particularly concerning motor insurance, the disclosure of infringements, driving offences, vehicles used for business purposes, modifications made to vehicles, and details of primary and additional drivers are deemed critical, as these factors significantly influence the calculation of insurance premiums (ICNZ, 2023).

Leveraging Predictive and Prescriptive Analysis would be the most suitable approach to comprehensively examine these core nominal, interval and ratio variables like 'age, gender, ethnicity, vehicle type, brand, daily km travelled, pending total ticket count, pending speeding tickets count and average driving speed, average working hours in a day' would uncover the

reasons behind speeding occurrences and define future trends. Through data visualization, it becomes effortless to identify areas of improvement and machine learning strategies to reduce speeding offences effectively. Traffic violation data like speeding, texting while driving, mobile phone usage while driving and not wearing seat belts must be promptly shared with the insurance provider as soon as it becomes available to the offender (Consumer NZ, 2023). Sit & Blake (2015) agree that this practice ensures complete transparency between the insurer and the buyer, allowing for effective risk mitigation for the insurer. Moreover, having access to up-to-date and readily available data when calculating insurance premiums is crucial (Consumer NZ, 2023). By adhering to this approach, insurers can make well-informed decisions on the premium to be quoted, and insurance buyers can benefit from fair and accurate premium assessments based on the most current and relevant information.

3.2 Problem 2: Seat belt Utilization – Manufacturing

Seat belts in vehicles have long been a critical safety measure for passengers, as their proper utilization ensures that passengers are positioned optimally during impacts, minimizing the risk of severe injuries in accidents and enhancing the effectiveness of airbags (Korgaonkar, 2022). While the seat belt utilization rate in New Zealand remains relatively high, it is concerning to note that non-seat belt utilization still contributed to 19%-30% of overall vehicle road deaths between 2006 and 2016, with a subsequent increase to 29-30% in 2017 (Mackie et al., 2017).

In response to life-threatening scenarios, car manufacturers have made significant strides in introducing manufacturing advancements to promote seat belt usage like Pretensioners, load limiters and Seat Belt Reminder Systems (SBR) (Clayton et al., 2006). These innovations prompt users with audio and visual cues to wear their seat belts every time they drive, ensuring enhanced safety measures on the road (Korgaonkar, 2022). This manufacturing breakthrough addresses security concerns and opens up a wealth of opportunities for car manufacturers to integrate advanced features for entertainment and other purposes, guiding in a transformative era in the automobile industry and the field of Information Technology (NHTSA, 2023). However, David Harkey, the president of the Insurance Institute for Highway Safety, has raised a noteworthy point of contention; he argues that while these audio and visual prompts are a step in the right direction, they may not be sufficiently loud and long-lasting to remind passengers to buckle up (KYMA, 2022). Harkey suggests that the prompts ideally last 90 seconds to ensure passengers' compliance with seat belt usage (KYMA, 2022). This critique

underscores the importance of continuously refining and optimizing such technologies to ensure their efficacy in bolstering road safety and safeguarding lives.

Extensive research has consistently revealed a concerning surge in casualty trends resulting from accidents (Mackie et al., 2017). Mackie et al., (2017) also state that given this alarming situation, conducting a thorough analysis and capitalizing on potential opportunities is imperative. The importance of addressing this challenge lies in adopting efficient data analytics methods for prompt and insightful analysis. In this regard, implementing Diagnostic and Predictive data analytics emerges as the most effective approach, allowing for proactive identification of potential risks and formulating strategic solutions to mitigate the impact of accidents and safeguard lives. As per Covaes et al., (2021), utilizing Regression Analysis is a valuable tool for establishing connections between two or more variables, such as ‘age, gender, ethnicity, license type, vehicle type and pending ticket count for not wearing seat belts, total ticket count, total pending ticket count’. This analytical approach enables the creation of meaningful relationships, allowing for comprehensive and in-depth data analysis (Covaes et al., 2021).

3.3 Problem 3: Driver Distraction – Automobile Technology Inventor (Research & Development)

Even a momentary diversion in a driver's attention for a duration exceeding two seconds can significantly escalate the likelihood of accidents (NZ Transport Agency, 2023). According to research by TeletracNavman (2023), distractions are categorized into manual, visual, and cognitive. Manual distraction can be understood when a driver takes one or both hands off the wheel, such as drinking or eating while driving (TeletracNavman, 2023). As per TeletracNavman (2023), Visual distraction involves taking one's eyes off the road, for instance, to read driving directions. Cognitive distraction, on the other hand, encompasses instances where a driver's mental focus is not on the road, as exemplified by using mobile phones while driving (TeletracNavman, 2023). Understanding these distinct types of distractions is vital in raising awareness and formulating effective strategies to promote safer driving habits and minimize the risks associated with driver inattention.

The continual evolution of technology within the automotive sector holds a pivotal position in addressing the issue of driver distraction. At the forefront of this transformative progress is the deployment of the Advanced Driver Assistance System (ADAS), a state-of-the-art automated

technology meticulously designed to counteract the very human errors and distractions that have proven to be catalysts for accidents (Salehian, 2022). The incorporation of ADAS aligns seamlessly with the New Zealand Road to ZERO initiative, a strategic framework meticulously designed to propel road safety endeavours and a substantially 40% reduction in severe injuries and fatalities (Ministry of Transport NZ, 2023). This technology uses an efficient system of sensors and cameras to identify driver errors and navigates obstacles, executing responsive actions to promptly rectify and address such occurrences (MBN, 2022). According to a report by Forbes in 2022, serious concerns and vulnerabilities within the ADAS safety system have come to light, primarily due to collisions involving various vehicles and cyclists, which would be resolved with time.

For a more profound understanding of human error, it is imperative to analyze the data diligently, unveiling intricate patterns and establishing comprehensive relationships. To understand the data, variables like 'age, gender, ethnicity, total tickets issued for mobile usage, total tickets issued for entry into a bus lane, vehicle manufacturer, vehicle type and manufacturing year of car, annual income, and human observatory variables from ADAS (observing driving patterns)' need to be analyzed to understand the percentage of cars which are ADAS enabled and its driver behaviour. Drivers might get distracted while on the road, yet they escape without receiving a ticket; this poses a significant issue that demands prompt consideration (Ashford et al., 2021). Leveraging Prescriptive analysis would facilitate a comprehensive grasp of the technological landscape and offer insights into enhancements recommended by machine learning, thereby ensuring the enhanced road safety of individuals.

4. Conclusion

In conclusion, this comprehensive exploration of general public behaviour and attitudes towards road safety underscores the critical importance of data-driven insights in shaping a safer and more secure road environment. The analysis has highlighted the intricate interplay between data measurement levels and analytics techniques, providing a holistic framework for understanding and addressing these challenges.

The application of diverse data analytics methods, spanning descriptive, diagnostic, predictive, and prescriptive approaches, offers a multifaceted toolkit to unearth the complexities of speeding, seat belt utilization, and driver distraction. By extracting valuable insights from these

analyses, decision-makers are empowered to craft well-informed and strategic interventions, contributing to safer roads and diminished accident rates. This collective effort can drive positive change in road safety by leveraging the power of data analytics.

5. Reference